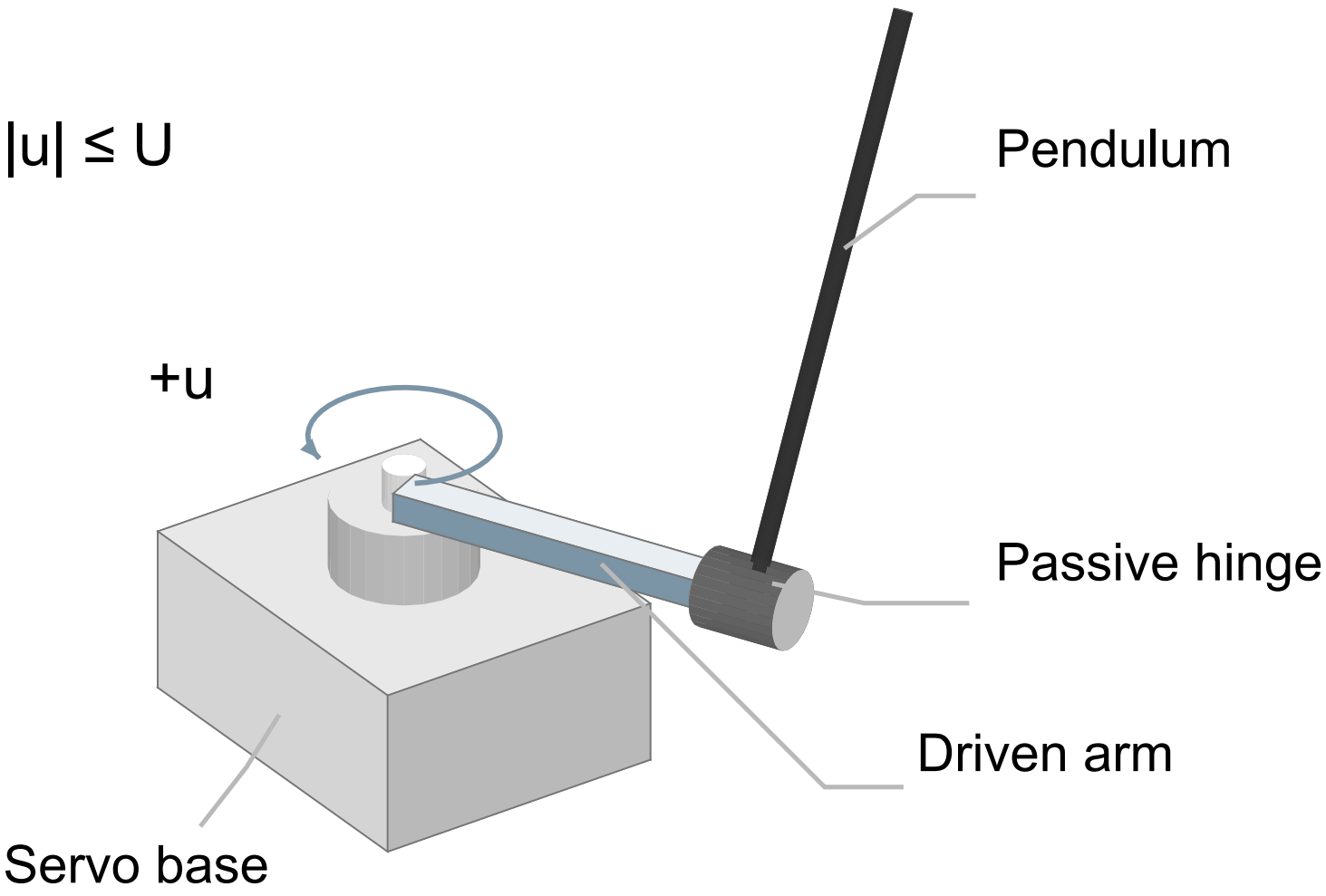
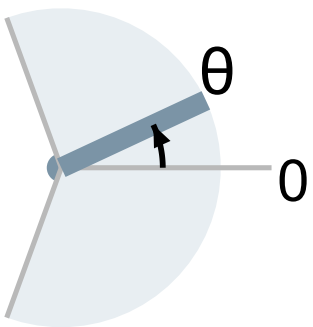
a Apparatus

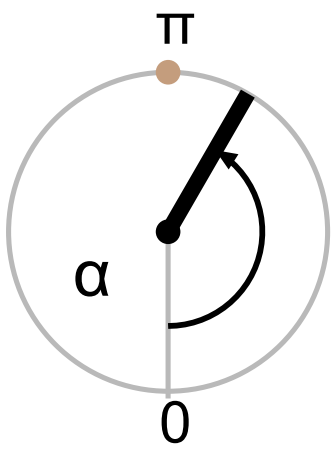


b Top



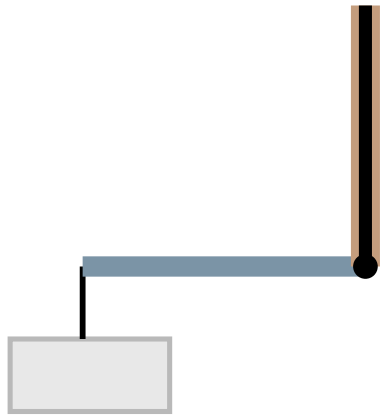
$|\theta| < \theta_{max}$

c Side



Free rotation

d Front



β = 0

ω = v = 0